

Tier 1: Central Server Coordinator (Flower ServerApp)

Global Model Registry

- Canonical parameters $\mathbf{w}_t$
- Global evaluation & checkpointing

Weights $\mathbf{w}_t$

Client Manager & Selector

- Samples active cohort $S_t \subset \{1, \dots, K\}$
- Tracks client availability & latency

Sampled cohort S_t

Federated Aggregator Strategy

- Federated averaging (FedAvg / FedMuon / FedProx)
- $\mathbf{w}_{t+1} = \mathbf{w}_t + \sum_k (|D_k| / |D|) \cdot \Delta \mathbf{w}_t^k$

[Downlink] Broadcast global weights $\mathbf{w}_t$ Deliver updates for aggregation

Tier 2: Secure Transport Channel (gRPC / TLS)

Model parameters only • Raw user data never leaves edge devices (Zero Data Leakage)

Receive $\mathbf{w}_t$

[Uplink] Transmit delta $\Delta \mathbf{w}_t^k$

Tier 3: Federated Edge Client $k \in \{1, \dots, K\}$ (Flower ClientApp)

Private Local Dataset D_k

- Strictly on-device storage
- No external access

Mini-batches

Local JAX/Flax Engine

- Forward loss & autodiff gradients
- Optax state update (AdamW / SGD)

Per-sample gradients

Privacy & Sparsification

- DP-SGD (Clipping C + Noise σ)
- Top-K sparsity + INT8 quantization

Processed delta $\Delta \mathbf{w}_t^k$

ClientApp Interface

- Ingests global weights $\mathbf{w}_t$
- Encodes parameter delta $\Delta \mathbf{w}_t^k$